(T) Roll No. .....

PAPER ID—21134

B.Tech. EXAMINATION, 2025

(Fourth Semester)

COMPUTER SCIENCE AND ENGINEERING

Operating System

Time : 3 *Hours* *Maximum Marks* : 70

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Unit. Q. No. 1 is compulsory. All questions carry equal marks.

1. (a) Define a process in the context of operating systems.

(b) What is the purpose of a Process Control Block (PCB) ?

(c) Explain the concept of safe and unsafe states in the context of deadlock prevention.

(d) Discuss one advantage and one disadvantage of preemptive scheduling.

(e) Explain the purpose of the superblock in a file system.

(f) Explain the concept of demand paging.

(g) Explain the concept of fragmentation.

7×2=14

Unit I

2. (a) Suppose a system uses SRTF scheduling with three processes, P1, P2 and P3. Their arrival times are 0 ms, 2 ms, and 4 ms, respectively. The CPU burst times for P1, P2, and P3 are 10 ms, 5 ms, and 8 ms, respectively. Calculate the average waiting time using SRTF scheduling. 7

(b) Compare and contrast preemptive and non-preemptive scheduling algorithms. Provide examples of each type. 7

3. (a) Consider a system with three processes, P1, P2, and P3, with priorities 3, 5, and 2, respectively. If the processes arrive at the same time and have CPU burst times

of 8 ms, 5 ms, and 10 ms, respectively, calculate the average turn-around time using priority scheduling. 7

(b) Describe the principles of multilevel queue scheduling. How are processes categorized into different queues and how is CPU time allocated among them ? 7

Unit II

4. (a) Discuss the role of mutual exclusion in preventing race conditions and ensuring program correctness. 7

(b) Explain the concept of message passing as a method of inter-process communication. 7

5. (a) Explain each of the four necessary conditions for deadlock and provide examples to illustrate each condition. 7

(b) Discuss two solutions to the producer-consumer problem : using semaphores and using monitors. Compare and contrast these solutions. 7

Unit III

6. Compare and contrast segmentation and paging as memory management techniques in virtual memory systems. 14

7. Suppose a computer system employs demand paging with a page size of 4 KB. If a process requires 64 KB of memory and the system has 256 KB of physical memory, calculate the following : **14**

(a) The number of pages required to store the process.

(b) If the process accesses pages in the following order :

1, 2, 3, 4, 5, 1, 2, 3, 4, 5

calculate the page faults using the FIFO page replacement algorithm.

Unit IV

8. (a) Consider a disk with 1000 tracks numbered from 0 to 999. The disk arm is currently at track 150 and the disk queue contains the following requests :

82, 170, 396, 520, 675.

Using the SCAN disk scheduling algorithm, calculate the total head movement required to satisfy all requests. 7

(b) Explain the contiguous allocation method for disk space allocation. What are the advantages and disadvantages of contiguous allocation ? 7

9. Compare and contrast the file system structures and management approaches used in UNIX and Windows operating systems. What are the key similarities and differences between these systems ? 14

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